NORTHERN WOLVES AIR CONDITIONING

Mechanical Contractor | New York & New Jersey

SUBMITTAL
#003

REFRIGERATION PIPING INSULATION

Spec Section: 23 07 19 — HVAC Piping Insulation

Elastomeric Insulation — 1.5" Outdoor/Main + 1" Branch-to-AHU

TRANSMITTAL INFORMATION

Project Name:	[PROJECT NAME]	**Date:**	March 22, 2026
Project Location:	[PROJECT LOCATION]	**Submitted By:**	Northern Wolves AC
General Contractor:	[GC NAME]	**Attention:**	[ENGINEER / ARCHITECT]
Architect / Engineer:	[A/E FIRM]	**Submittal Type:**	Initial Submission

DESCRIPTION OF SUBMITTAL

Closed-cell elastomeric pipe insulation for refrigerant piping systems.
Outdoor piping & indoor runs to branch boxes: 1-1/2" wall thickness (EPDM based).
Branch boxes to air handlers: 1" wall thickness (NBR/PVC based).
Includes adhesive, insulation tape, and UV-protective finish for outdoor sections.

PRODUCT INFORMATION

Manufacturer:	Armacell LLC
Product:	AP ArmaFlex Black Tube Insulation
Outdoor / Main Runs:	1-1/2" wall — EPDM based, max 300°F
Branch to AHU:	1" wall — NBR/PVC based, max 220°F
Adhesive:	ArmaFlex 520 Adhesive — for seams and butt joints
Tape:	AP ArmaFlex Insulation Tape — for fittings
Outdoor Protection:	ArmaFlex WB Finish — UV/ozone protection
Fire Rating:	FHC 25/50 per ASTM E84, Plenum Rated
Spec Compliance:	ASTM C534 Type I Gr.1, NFPA 90A/90B, NYC MEA 107-89M

NOTES

- HIGHLIGHTED items on attached data sheets indicate submitted wall thicknesses.
- YELLOW = 1-1/2" wall (outdoor / main runs to branch boxes).
- ORANGE = 1" wall (branch boxes to air handlers).
- All outdoor ArmaFlex must be coated with ArmaFlex WB Finish for UV protection.

SOLUTIONS FOR ENERGY SAVINGS

AP ArmaFlex®

Black and White Tube Insulation

The original, fiber-free, flexible elastomeric pipe insulation for reliable protection against condensation and energy loss.

// Fiber-free, formaldehyde-free, low VOC and non-particulating formulation protects indoor air quality
// Closed-cell structure provides excellent condensation control
// Built-in vapor retarder eliminates need for additional vapor retarder
// Rated 25/50 for use in return air plenums

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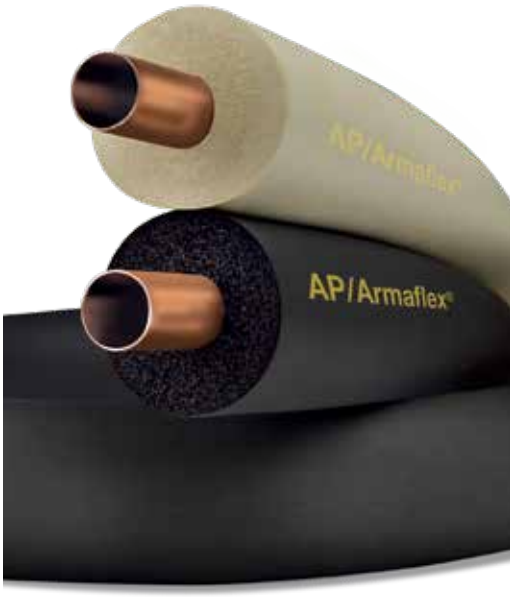

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TECHNICAL DATA – AP ArmaFlex® BLACK AND WHITE TUBE INSULATION

refrigeration pipe insulation

Description

Black or off-white flexible closed-cell elastomeric thermal insulation in a tubular form

Applications

Insulation for piping associated with HVAC, VRV and VRF systems, chillers, hot and cold water, refrigeration

Specification Compliance

ASTM C 534, Type I – Grade 1 ASTM D 1056, 2C1 ASTM E 84, UL723	ASTM G21/C1338 CAN/ULC S102 ① City of New York Approval MEA 107-89M	MIL-P-15280J, FORM T (black product only) ② NFPA 90A, 90B	UL 181 UL 94 5V-A, V-0, File E55798 (black product only)

Approvals, Certifications, Compliances

- 3rd party certified by FM Approvals through 1-1/2" wall thickness per FM 4924
- GREENGUARD® Children & Schools Indoor Air Quality certified.
- Manufactured without CFCs, HFCs, HCFCs, PBDEs, or Formaldehyde.
- Made with EPA registered Microban® antimicrobial product protection.
- All Armacell facilities in North America are ISO 9001 certified.
- Plenum Rated
- Conforms to ASHRAE 90.1 Energy Standards
- Conforms to building codes: International Mechanical Code, IMC, International Energy Conservation Code, IECC, International Residential Code, IRC, Title 24 California Building Energy Efficiency Standards.

Typical Properties

Specifications	BRANCH-TO-AHU (1" WALL) **3/8" through 1" Walls** (NBR/PVC based)	OUTDOOR/MAIN (1-1/2" WALL) **1-1/2" and 2" Walls** (EPDM based)	Test Method
Thermal Conductivity: Btu • in/h • ft2 • °F (W/mK)			
50°F Mean Temperature (10°C) 75°F Mean Temperature (24°C) 100°F Mean Temperature (38°C) 125°F Mean Temperature (52°C)	0.235 (0.034) 0.245 (0.0353) 0.257 (0.037) 0.268 (0.039)	0.278 (0.040) 0.28 (0.040) 0.289 (0.041) 0.300 (0.043)	ASTM C 177 or C 518
Water Vapor Permeability: Perm-in. [Kg/(s • m • Pa)]	0.05 (0.725 x 10^{-13})	0.08 (1.16 x 10^{-13})	ASTM E 96, Procedure A
Flame Spread and Smoke Developed Index:	25/50 rated	25/50 rated	ASTM E 84, UL 723 CAN/ULC S102 ①
Water Absorption, % by Volume:	0.2 %	0.2 %	ASTM C 209 or ASTM C1763
Mold Growth: **Fungi Resistance:**	Passed	Passed	UL181 ASTM G21/C1338
Maximum Service Temperature	220°F (105°C) ③	300°F (149°C) ④	ASTM C534
Minimum Service Temperature ⑤	-297°F (-183°C) ⑥	-297°F (-183°C) ⑥	ASTM C534

Sizes

Wall Thickness (nominal)	3/8", 1/2", 3/4", 1", 1-1/2", 2" (10, 13, 19, 25, 38, 50 mm)
Inside Diameter, Tubular	1/4" (black product only) ID to 10"ID (6 mm ID to 250 mm ID)
Length of Sections, Feet, Tubular	6' (1.8 m) (Some larger sizes may be shipped in two 3' sections)

Outdoor Use

Painting with WB Finish or other protective jacketing is required to prevent damage to the insulation in exterior applications and to comply with the insulation protection sections of the International Energy Conservation Code (IECC) and ASHRAE 90.1.

① AP ArmaFlex meets CAN/ULC S102 through 1" wall. AP ArmaFlex Black tested. AP ArmaFlex White determined to be comparable through 1" wall.
② AP ArmaFlex meets MIL-P-15280J through 1" wall.
③ AP ArmaFlex Pipe Insulation can withstand temperatures as high as 250°F for 96 hour time periods when tested according to ASTM C411 - Standard Test Method for Hot-Surface Performance of High-Temperature Thermal Insulation.
④ 1-1/2" and 2" AP ArmaFlex tubes are formulated with EPDM rubber giving them a higher upper use temperature than AP ArmaFlex tubes less than 1-1/2" wall thickness.
⑤ At temperatures below -20°F (-29°C), elastomeric insulation starts to become less flexible. However, this characteristic does not affect thermal efficiency and resistance to water vapor permeability of ArmaFlex insulation.
⑥ For applications of -40°F to -297°F (-40°C to -183°C), contact Armacell.

GREENGUARD Certified products are certified to GREENGUARD standards for low chemical emissions into indoor air during product usage. For more information, visit ul.com/gg.

Microban antimicrobial product protection is limited to the product itself and is not designed to protect the users of these products from disease causing microorganisms, or as a substitute for normal cleaning and hygiene practices. Microban International, Ltd. makes neither direct nor implied health claims for the products containing Microban® antimicrobial product protection. Data, photomicrographs and information presented are based on standard laboratory tests and are provided for comparative purposes to substantiate antimicrobial activity for non-public health uses. Microban is a registered trademark of Microban International, Ltd.

AP ArmaFlex BLACK TUBES - R VALUES

3/8" WALLS	Nominal Copper or IPS Size	R-Value
IPAPT01438	3/16" Copper	2.8
IPAPT03838	1/4" Copper	2.8
IPAPT01238	3/8" Copper	2.6
IPAPT05838	1/2" Copper	2.6
IPAPT03438	5/8" Copper	2.4
IPAPT07838	3/4" Copper	2.4
IPAPT11838	1" Copper	2.3
IPAPT13838	1-1/4" Copper	2.2
IPAPT15838	1-1/2" Copper	2.5
IPAPT11038	1-1/2" IPS	2.4
IPAPT21838	2" Copper	2.4
IPAPT20038	2" IPS	2.4
IPAPT25838	2-1/2" Copper	2.4
IPAPT21038	2-1/2" IPS	2.4
IPAPT31838	3" Copper	2.4
IPAPT30038	3" IPS	2.3

1/2" WALLS	Nominal Copper or IPS Size	R-Value
IPAPT01412	3/16" Copper	3.8
IPAPT03812	1/4" Copper	3.3
IPAPT01212	3/8" Copper	3.3
IPAPT05812	1/2" Copper	3.4
IPAPT03412	5/8" Copper	3.3
IPAPT07812	3/4" Copper	3.3
IPAPT11812	1" Copper	3.3
IPAPT13812	1-1/4" Copper	3.2
IPAPT15812	1-1/2" Copper	3.2
IPAPT11012	1-1/2" IPS	3.1
IPAPT21812	2" Copper	3.2
IPAPT20012	2" IPS	3.2
IPAPT25812	2-1/2" Copper	3.2
IPAPT21012	2-1/2" IPS	3.2
IPAPT31812	3" Copper	3.2
IPAPT30012	3" IPS	3.1
IPAPT35812	3-1/2" Copper	3.1
IPAPT41812R	4" Copper	3.1
IPAPT40012R	4" IPS	3.0
IPAPT50012R	5" IPS	3.0
IPAPT60012R	6" IPS	3.0
IPAPT80012R	8" IPS	2.9

3/4" WALLS	Nominal Copper or IPS Size	R-Value
IPAPT01434	3/16" Copper	6.4
IPAPT03834	1/4" Copper	5.9
IPAPT01234	3/8" Copper	5.5
IPAPT05834	1/2" Copper	5.6
IPAPT03434	5/8" Copper	5.5
IPAPT07834	3/4" Copper	5.4
IPAPT11834	1" Copper	5.4
IPAPT13834	1-1/4" Copper	5.3
IPAPT15834	1-1/2" Copper	5.1
IPAPT11034	1-1/2" IPS	4.9
IPAPT21834	2" Copper	4.8
IPAPT20034	2" IPS	5.2
IPAPT25834	2-1/2" Copper	4.7
IPAPT21034	2-1/2" IPS	5.0

3/4" WALLS (continued)	Nominal Copper or IPS Size	R-Value
IPAPT31834	3" Copper	4.6
IPAPT30034	3" IPS	4.9
IPAPT35834	3-1/2" Copper	4.5
IPAPT41834R	4" Copper	4.5
IPAPT40034R	4" IPS	4.8
IPAPT50034R	5" IPS	4.7
IPAPT60034R	6" IPS	4.6
IPAPT80034R	8" IPS	4.5

1" WALL — BRANCH TO AHU

1" WALLS	or IPS Size	R-Value
IPAPT01410	3/16" Copper	8.3
IPAPT03810	1/4" Copper	7.3
IPAPT01210	3/8" Copper	7.2
IPAPT05810	1/2" Copper	7.2
IPAPT03410	5/8" Copper	7.0
IPAPT07810	3/4" Copper	7.0
IPAPT11810	1" Copper	7.2
IPAPT13810	1-1/4" Copper	7.2
IPAPT15810	1-1/2" Copper	7.2
IPAPT11010	1-1/2" IPS	6.9
IPAPT21810	2" Copper	6.8
IPAPT20010	2" IPS	7.1
IPAPT25810	2-1/2" Copper	6.5
IPAPT21010	2-1/2" IPS	6.8
IPAPT31810	3" Copper	6.3
IPAPT30010	3" IPS	6.6
IPAPT35810	3-1/2" Copper	6.2
IPAPT41810R	4" Copper	6.1
IPAPT40010R	4" IPS	6.4
IPAPT50010R	5" IPS	6.2
IPAPT60010R	6" IPS	6.1
IPAPT80010R	8" IPS	5.9
IPAPT10010R	10" IPS	5.8

1-1/2" WALL — OUTDOOR / TO BRANCH

1-1/2" WALLS	or IPS Size	R-Value
IPAPT03815	1/4" Copper	13.7
IPAPT01215	3/8" Copper	12.7
IPAPT05815	1/2" Copper	12.0
IPAPT03415	5/8" Copper	11.3
IPAPT07815	3/4" Copper	10.8
IPAPT11815	1" Copper	10.1
IPAPT13815	1-1/4" Copper	9.6
IPAPT15815	1-1/2" Copper	9.2
IPAPT11015	1-1/2" IPS	8.7
IPAPT21815	2" Copper	8.6
IPAPT20015	2" IPS	8.8
IPAPT25815	2-1/2" Copper	8.2
IPAPT21015	2-1/2" IPS	8.4
IPAPT31815	3" Copper	7.9
IPAPT30015	3" IPS	8.1
IPAPT35815	3-1/2" Copper	7.7
IPAPT41815R	4" Copper	7.5
IPAPT40015R	4" IPS	7.8
IPAPT50015R	5" IPS	7.5
IPAPT61815R	6" Copper	7.4
IPAPT60015R	6" IPS	7.3

1-1/2" WALL (continued)

1-1/2" WALLS (continued)	Nominal Copper or IPS Size	R-Value
IPAPT80015R	8" IPS	7.0
IPAPT10015R	10" IPS	6.8

2" WALLS	Nominal Copper or IPS Size	R-Value
IPAPT03820	1/4" Copper	19.7
IPAPT01220	3/8" Copper	18.2
IPAPT05820	1/2" Copper	17.2
IPAPT03420	5/8" Copper	16.2
IPAPT07820	3/4" Copper	15.5
IPAPT11820	1" Copper	14.5
IPAPT13820	1-1/4" Copper	13.7
IPAPT15820	1-1/2" Copper	13.1
IPAPT11020	1-1/2" IPS	12.4
IPAPT21820	2" Copper	12.2
IPAPT20020	2" IPS	12.3
IPAPT25820	2-1/2" Copper	11.6
IPAPT21020	2-1/2" IPS	11.7
IPAPT31820	3" Copper	11.1
IPAPT30020	3" IPS	11.2
IPAPT35820	3-1/2" Copper	10.7
IPAPT41820R	4" Copper	10.5
IPAPT40020R	4" IPS	10.7
IPAPT50020R	5" IPS	10.2
IPAPT60020R	6" IPS	9.9
IPAPT80020R	8" IPS	9.5
IPAPT10020R	10" IPS	9.2

*These specifications are based on the measurement methods employed by Armacell. Other methods may not result in the same values and cannot be used to determine if the product is within the given tolerance.

Description

An air-drying contact adhesive that is excellent for joining seams and butt joints of ArmaFlex Pipe and Sheet Insulations

Specification Compliance

MIL-A-24179A Type II, Class 1	ASTM G21/C1338

Approvals, Certifications, Compliances

- All Armacell facilities in North America are ISO 9001 certified.

Typical Properties

Specifications	Values	
Color	Greenish yellow	
Net Weight	6.9 pounds per gallon (828 g/l)	
Composition	Synthetic rubber base with synthetic resins and fillers added; hydrocarbon- and ketone-type solvents.	
Volatile Organic Compounds (VOC) Content	615 g/l	
Solids Content	Approximately 19% by weight	
Coverage	200 sq ft (5 m^2/l) per gallon max, single coat (depending upon porosity of materials bonded and air temperature)	
Shelf Life	1-1/2 years in original sealed container; Store in a well ventilated area with storage temperature 60° F to 80° F (16° C to 27° C).	
Temperature Limits	250°F (120°C) — ArmaFlex Pipe Insulation seams and joints 180°F (82°C) — Full bonding sheet insulation	
Tack Time	1–5 minutes depending on ambient conditions	
Bond Time	Immediate	
Full Cure Time	36 hours	
Container Sizes	Half-pint and pint brush-top cans and pint, quart, and gallon containers	
Wet Flash Point	Below 20°F (-7°C) (TOC)	
Flame Spread and Smoke Developed Index	25/50 rated	ASTM E 84
Minimum Application Temperature	40°F (4°C)	

All data and technical information are based on results achieved under typical application conditions. It is the customer's responsibility to verify if the product is suitable for the intended application. The responsibility for professional and correct installation and compliance with relevant building regulations lies with the customer. By ordering/receiving product you accept the **Armacell General Terms and Conditions of Sale** applicable in the region. Please request a copy if you have not received these.

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ArmaFlex | 520 Adhesive TDS | 082020 | NA| EN-A | 083

ABOUT ARMACELL

As the inventors of flexible foam for equipment insulation and a leading provider of engineered foams, Armacell develops innovative and safe thermal, acoustic and mechanical solutions that create sustainable value for its customers. Armacell's products significantly contribute to global energy efficiency making a difference around the world every day. With 3100 employees and 24 production plants in 16 countries, the company operates two main businesses, Advanced Insulation and Engineered Foams. Armacell focuses on insulation materials for technical equipment, high-performance foams for high-tech and lightweight applications and next generation aerogel blanket technology.

For more information, please visit:
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SOLUTIONS FOR ENERGY SAVINGS

AP ArmaFlex®

Insulation Tape

The mold resistant, elastomeric, AP ArmaFlex pressure sensitive insulation tape that provides a fast, easy method of insulating pipes and fittings

// High-quality AP ArmaFlex insulation
// Fast on difficult pipes and fittings
// Controls condensation and heat loss
// Clean, smooth appearance

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TECHNICAL DATA – AP ArmaFlex® INSULATION TAPE

Description

A black, closed cell, mold resistant, elastomeric thermal insulation tape for insulating pipes and fittings

Specification Compliance

ASTM C 534, Type II – Grade 1 ASTM E 84, UL723 CAN/ULC S102	NFPA 90A, 90B UL 181 ASTM G-21/C1338	ASTM D 1056, 2C1 Grade SBE 3 City of New York approval MEA 96-85-M

Approvals, Certifications, Compliances

- Manufactured without CFCs, HFCs, HCFCs, PBDEs, or Formaldehyde.
- Made with EPA registered Microban® antimicrobial product protection.
- All Armacell facilities in North America are ISO 9001 certified.

Typical Properties

Specifications	Values	Test Method
Thermal Conductivity: Btu • in/h • ft2 • °F (W/mK)		
75°F Mean Temperature (24°C) 100°F Mean Temperature (38°C)	0.25 (0.036) 0.257 (0.037)	ASTM C 177 or C 518
Water Vapor Permeability: Perm-in. [Kg/(s • m • Pa)]	0.05 (0.725 x 10^{-13})	ASTM E 96, Procedure A
Flame Spread and Smoke Developed Index:	25/50 rated	ASTM E 84, UL 723 CAN/ULC S102
Mold Growth: **Fungi Resistance:**	Pass Pass	UL181 ASTM G21/C1338
Maximum Service Temperature	Up to 180°F (82°C)	ASTM C534
Minimum Service Temperature	-30°F (-34°C)	ASTM C534

Microban antimicrobial product protection is limited to the product itself and is not designed to protect the users of these products from disease causing microorganisms, or as a substitute for normal cleaning and hygiene practices. Microban International, Ltd. makes neither direct nor implied health claims for the products containing Microban® antimicrobial product protection. Data, photomicrographs and information presented are based on standard laboratory tests and are provided for comparative purposes to substantiate antimicrobial activity for non-public health uses. Microban is a registered trademark of Microban International, Ltd.

All data and technical information are based on results achieved under typical application conditions. It is the customer's responsibility to verify if the product is suitable for the intended application. The responsibility for professional and correct installation and compliance with relevant building regulations lies with the customer. By ordering/receiving product you accept the **Armacell General Terms and Conditions of Sale** applicable in the region. Please request a copy if you have not received these.

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PROTECTION INDOORS AND OUT

ArmaFlex WB Finish

ArmaFlex® WB Finish is the durable, white, protective finish formulated for use over all ArmaFlex insulations for protection against ultraviolet radiation, weather and ozone damage or where a clean, white aesthetic is desired.

// Suitable for both indoor and outdoor applications.
// Provides a clean, white, protective finish which guards against UV and ozone damage.
// Formulated for cold weather flexibility to resist cracking.
// Water-based, 100% acrylic is fast drying without solvent fumes. Cleans up with soap and water.
// Outdoor surfaces should be recoated every 2-4 years.

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TECHNICAL DATA – ARMAFLEX® WB FINISH

Description

White, water-based, 100% acrylic coating

Typical Properties

Physical Properties	Value
Color:	Available in Standard White. Tinting possible for indoor use only. No tinting for outdoor applications.
Net Weight:	11.3 lbs. per gallon (1.35 kg/l)
Composition:	Pigmented latex V.O.C. Less than 50 g/l (calculated)
Solid Content:	Approximately 65% by weight and 53% by volume.
Viscosity:	95 - 120 KU
Application:	Brush or roller. Apply at temperatures above 50°F (10°C) and when no freezing temperatures are expected for 24 hours
Coverage:	200 - 400 sq. ft. (37 sq.m.) per gallon per single brush coat, depending on temperature and substrate.
Shelf Life:	One year in original sealed container. Storage temperature 60° to 80°F (16° to 27°C). **Keep from freezing.**
Minimum Drying Time:	Four hours minimum for application of second coat, at least 24 hours to dry completely
Container Sizes:	One-gallon or quart containers
Flammability:	Not flammable; water-based
Surface-Burning Characteristics:	Dry: ASTM E 84 Method Flame-Spread Index 0 Smoke Developed Index 0

Outdoor Pipe Insulation Finish Paint

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WB Finish | ArmaFlex | TDS | 112019 | NA | EN-A | 082

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